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Q1) What is data, and why is it important in the modern world?

Ans:- Data is **raw information** that helps in making decisions. It can be numbers, text, images, or even videos. data may not have a clear meaning, but when it is organized, analyzed, and interpreted, it becomes useful information that supports understanding and decision-making.

Importance of data in the modern world:-

- **Helps in Decision Making :-** Companies use data to identify trends and make strategic decisions.
- **Identifies Patterns and Trends :-** Helps in understanding customer behavior and market trends.
- **Automates Processes :-** Businesses use data to streamline operations and reduce manual effort.
- **Education Enhancement :-** Educational institutions use data to evaluate student performance, identify learning gaps, and improve teaching methods.

Real-Life Example

Think about an **online shopping website**. Every time you purchase something, your buying patterns are recorded. This helps the website suggest products based on your interests.

Q2) What is the difference between qualitative and quantitative data?

Ans:- Data is classified into two main types: Qualitative Data and Quantitative Data. Both types are used for analysis, but they differ in their nature and purpose.

	Qualitative Data	Quantitative Data
1	Qualitative data describes the qualities or characteristics of an object or person.	Quantitative data represents numerical values that can be measured or counted
2	It is non-numerical in nature.	It is numerical in nature.
4	It answers questions such as what type, which category, or what kind.	It answers questions such as how many, how much, or how often.
5	It cannot be measured using mathematical calculations.	It can be measured and analyzed using mathematical and statistical methods.
6	Examples include color, gender, nationality, customer feedback, and product category.	Examples include age, height, weight, salary, marks, temperature, and number of students.
7	It is mainly used to understand opinions, behaviors, and characteristics.	It is mainly used to measure quantities and identify trends or patterns.
8	Example :- customer feedback, interview transcripts, colors, or product reviews.	Example:- Age: 22 years Height: 170 cm Weight: 65 kg

Q3) What is data analysis, and how does it differ from data science?

Ans :- Data Analysis means converting raw data into useful knowledge through structured steps. The primary purpose of data analysis is to convert raw data into useful information that helps individuals and organizations make informed decisions

Steps Involved:

1. **Collecting Data**
From spreadsheets, surveys, databases, etc.
2. **Cleaning Data**
Fixing errors and filling missing values.
3. **Analyzing Data**
Using basic statistics to discover trends and patterns.
4. **Visualizing Data**
Presenting findings with charts and dashboards.

Real-World Example:

Amazon uses customer feedback to improve its products.

Data Analysis is Differ from Data Science because:-

Data science is a broader field that includes data analysis along with programming, statistics, machine learning, artificial intelligence, and big data technologies. In addition to analyzing existing data, data science focuses on building predictive models and intelligent systems that can forecast future outcomes and automate decision-making. It answers questions such as "What is likely to happen?" and "What actions should be taken?"

Another key difference is that data analysts mainly use tools such as Microsoft Excel, SQL, Power BI, Tableau, and Python to create reports, dashboards, and visualizations. Data scientists use advanced programming languages and machine learning frameworks, such as Python, R, TensorFlow, and Scikit-learn, to develop predictive models and AI-based solutions.

Q4) What are the key steps involved in a typical data analysis process?

Ans:- Data analysis is a systematic process of collecting, organizing, examining, and interpreting data to extract meaningful insights and support decision-making. The following are the key steps involved in a typical data analysis process:

1. Define the Problem or Objective

The first step is to clearly identify the problem or business objective. This helps determine what information is needed and what questions the analysis should answer. A well-defined objective ensures that the analysis remains focused and relevant.

2. Collect Data

The next step is to gather data from appropriate sources such as databases, surveys, websites, sensors, business records, or online applications. The collected data should be accurate, relevant, and sufficient for the analysis.

3. Clean and Prepare the Data

Raw data often contains missing values, duplicate records, incorrect entries, or inconsistencies. Data cleaning involves removing errors, handling missing values, correcting inconsistencies, and transforming the data into a suitable format for analysis.

4. Explore and Analyze the Data

Once the data is prepared, it is explored using statistical methods, charts, graphs, and visualization tools to identify patterns, relationships, trends, and anomalies. This step helps answer the research or business questions.

5. Interpret the Results

After completing the analysis, the results are interpreted to understand their meaning and significance. The findings are compared with the original objectives to determine whether the problem has been solved or useful insights have been obtained.

6. Present and Communicate the Findings

The analyzed results are presented in the form of reports, dashboards, charts, graphs, or presentations. Clear communication of findings helps managers, stakeholders, and decision-makers understand the results and take appropriate actions.

7. Make Decisions and Take Action

The final step is to use the insights gained from the analysis to make informed decisions, improve business processes, solve problems, or develop future strategies. Organizations often monitor the outcomes and repeat the analysis when new data becomes available.

5) What tools and techniques are commonly used in data science?

Ans:- Data science uses different tools and methods to gather, organize, analyze, and present data. These help data scientists find useful information and make better decisions based on data.

Commonly Used Tools

1. Python:- Python is the most popular programming language in data science. It is used for data analysis, machine learning, automation, and data visualization because of its simple syntax and powerful libraries.
2. R:- R is mainly used for statistical analysis and creating graphs. It is widely used in research and academic projects.
3. SQL (Structured Query Language):- SQL is used to store, retrieve, update, and manage data in databases. It helps in handling large amounts of structured data.
4. Microsoft Excel:- Excel is used for data entry, data cleaning, calculations, and creating charts. It is commonly used for basic data analysis.
5. Power BI and Tableau:- These are data visualization tools that help create interactive dashboards, reports, and charts, making data easy to understand.

Commonly Used Techniques

1. Data Collection:- Collecting data from various sources such as databases, websites, surveys, and sensors.
2. Data Cleaning:- Removing missing values, duplicate records, and errors to improve data quality.
3. Data Visualization:- Representing data using graphs, charts, and dashboards to identify patterns and trends.
4. Statistical Analysis:- Applying statistical methods to summarize data, identify relationships, and support decision-making.
5. Machine Learning:- Developing models that can learn from data and make predictions.